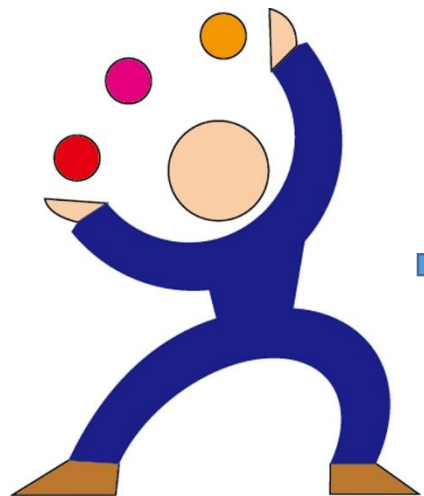
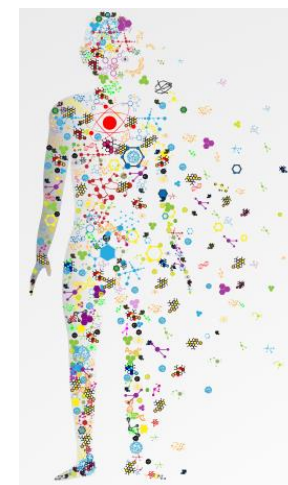


生物信息学

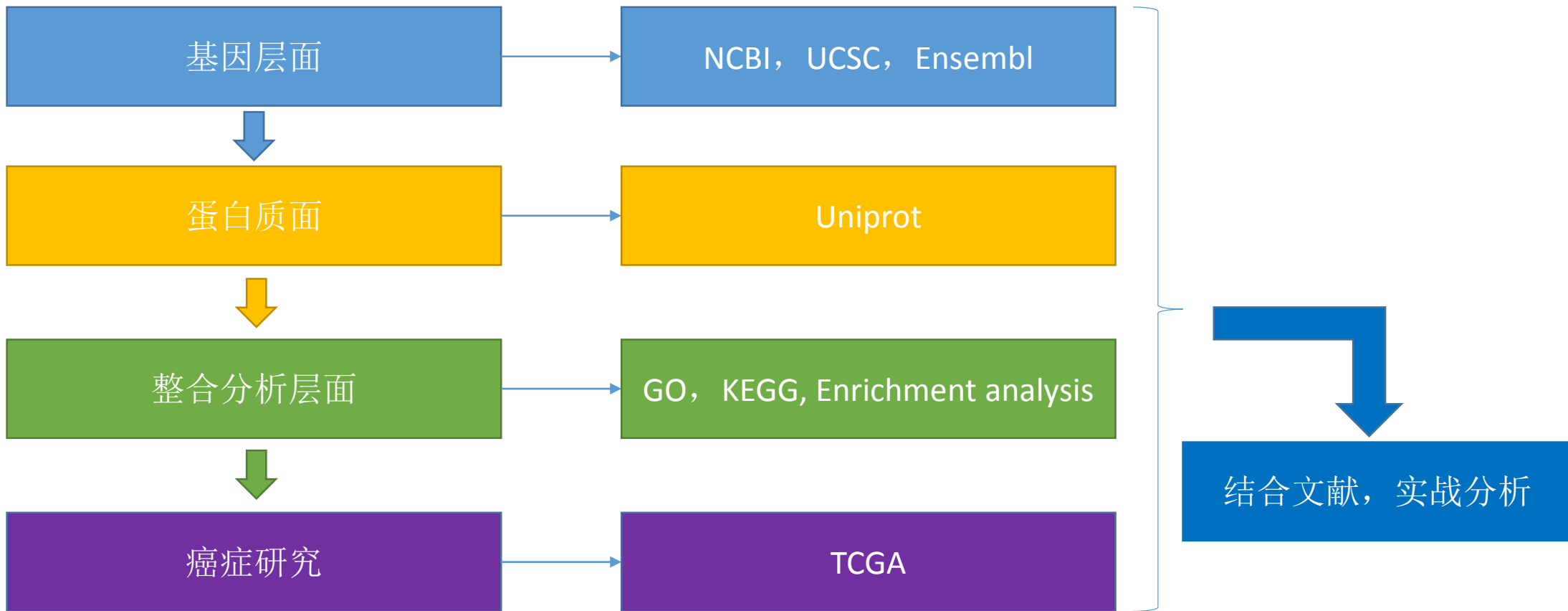


Helix Life 解螺旋

Redocn.com



课程安排



第一堂课内容提要

- 1. NCBI及GenBank的介绍
- 2. 如何检索
- 3. 解释检索结果
- 4. 下载选定结果
- 4. NCBI其他资源介绍

NCBI简介



- <http://www.ncbi.nlm.nih.gov/>
- National Center for Biotechnology Information
- 1988年11月4日建立
- 使命：
 - 建立关于分子生物学、生物化学、和遗传学知识的存储、分析的自动系统
 - 实现 关于用于分析生物学重要分子和复合物的结构和功能的基于计算机的信息处理的、先进方法的研究
 - 加速生物技术研究者和医药治疗人员对数据库和软件的使用。
 - 全世界范围内的生物技术信息收集的合作努力。

National Center for
Biotechnology Information

All Databases ▾

Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

[About the NCBI](#) | [Mission](#) | [Organization](#) | [NCBI News](#) | [Blog](#)

Submit

Deposit data or manuscripts
into NCBI databases



Download

Transfer NCBI data to your
computer



Learn

Find help documents, attend a
class or watch a tutorial



Develop

Use NCBI APIs and code
libraries to build applications



Analyze

Identify an NCBI tool for your
data analysis task



Research

Explore NCBI research and
collaborative projects



Popular Resources

[PubMed](#)[Bookshelf](#)[PubMed Central](#)[PubMed Health](#)[BLAST](#)[Nucleotide](#)[Genome](#)[SNP](#)[Gene](#)[Protein](#)[PubChem](#)

NCBI Announcements

June 15th webinar: Using NCBI
Resources and Variant Interpretation
Tools for the Clinical Community

01 Jun 2016

In two weeks, NCBI will present a

Epigenomics database to be retired June
1, 2016

27 May 2016

The Epigenomics database, a public
repository that was developed to archive

NCBI launches new Twitter account for
NCBI Bookshelf

23 May 2016

NCBI has a new Twitter feed -
@ncbihooks - to announce new books

[More...](#)

<http://www.ncbi.nlm.nih.gov/>

螺旋

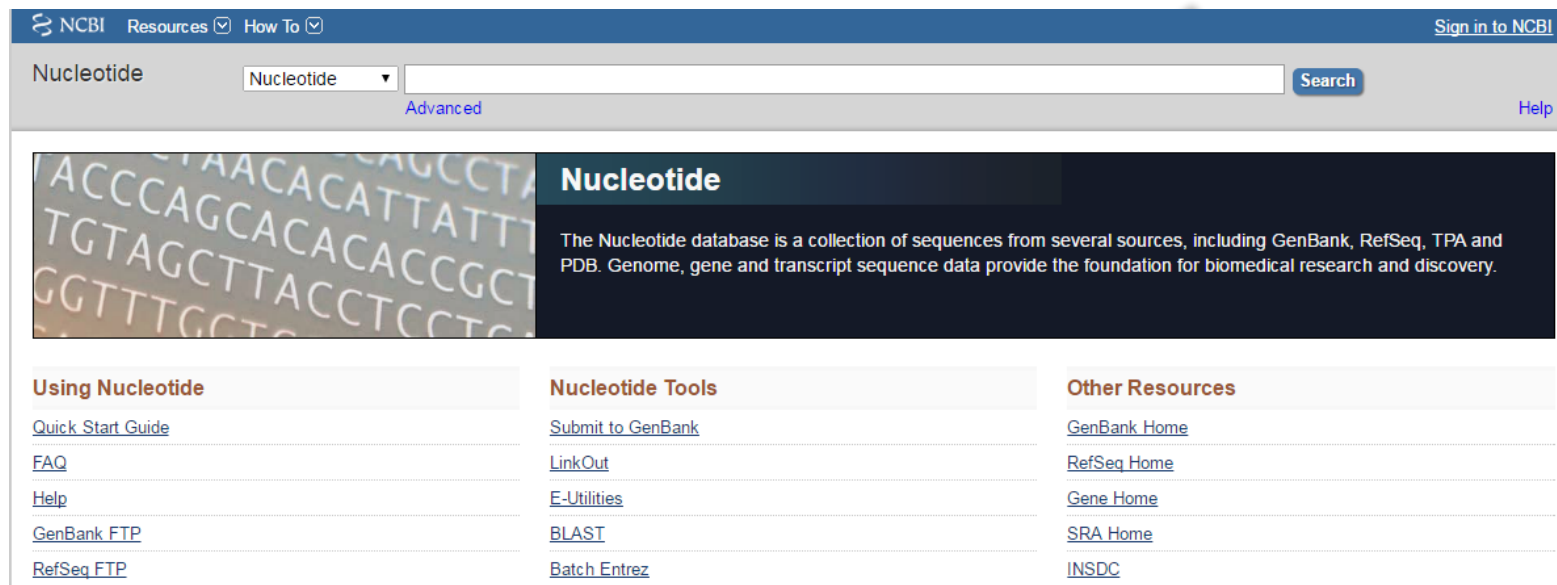
NCBI的发展历程

- 首先创建了GenBank数据库
- 1991年开发了Entrez 数据库检索系统。

该系统整合了GenBank、EMBL、PIR和SWISS-PROT等数据库的序列信息以及MEDLINE有关序列的文献信息，并通过相关链接，将他们有机地结合在一起。
- NCBI还提供了其它数据库：
 - 在线人类孟德尔遗传（OMIM）
 - 三维蛋白结构的分子模型数据库（MMDB）
 - 人类基因序列集成（UniGene）
 - 人类基因组基因图谱（GMHG）
 - 生物门类（Taxonomy）
 - ...

GenBank

- NIH遗传序列数据库



- 收集并注释了所有可以公开获得的DNA序列
- 国际核酸序列数据库：
 - GenBank
 - 日本DNA数据库（DDBJ）
 - 欧洲分子生物学实验室核算数据库（EMBL）这三个组织每天交换数据。
- 每两个月会有新的release, 每18个月其包含的碱基数会翻一倍。

GENBANK AND WGS STATISTICS



		GenBank		WGS	
Release	Date	Bases	Sequences	Bases	Sequences
3	Dec 1982	680338	606		
14	Nov 1983	2274029	2427		
20	May 1984	3002088	3665		

129	Apr 2002	19072679701	16769983	692266338	172768
130	Jun 2002	20648748345	17471130	3267608441	397502
131	Aug 2002	22616937182	18197119	3848375582	427771

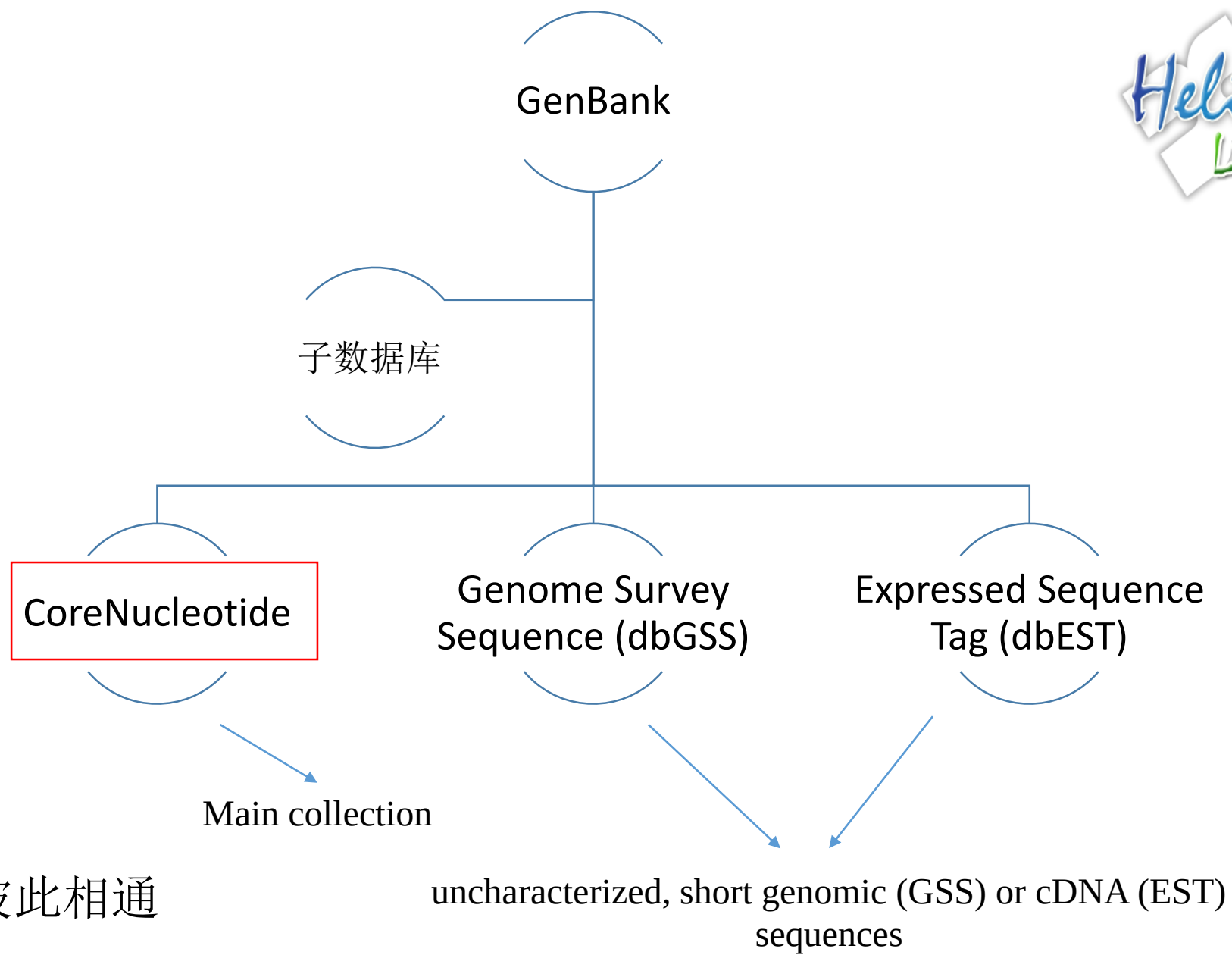
211	Dec 2015	203939111071	189232925	1297865618365	317122157
212	Feb 2016	207018196067	190250235	1399865495608	333012760
213	Apr 2016	211423912047	193739511	1452207704949	338922537

现有序列数: 193,739,511 (traditional division)
338,922,537 (WGS division)

$$\frac{193,739,511}{606} = 319,702 !!!$$

GenBank 数据来源

- 1) 直接来源于测序工作者提交的序列
- 2) 与其他数据机构协作交换
- 3) 美国专利局提供的数据库



搜索结果彼此相通

整节课的问题

- 例如我对一个基因 **BRCA1**感兴趣，想知道：
 - 1) 基因在哪条染色体上？
 - 2) 该基因的功能是什么？
 - 3) 该基因的DNA序列是什么？ Protein序列是什么？

第一堂课内容提要

- 1. NCBI及GenBank的介绍
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- 4. 下载选定结果
- 4. NCBI其他资源介绍

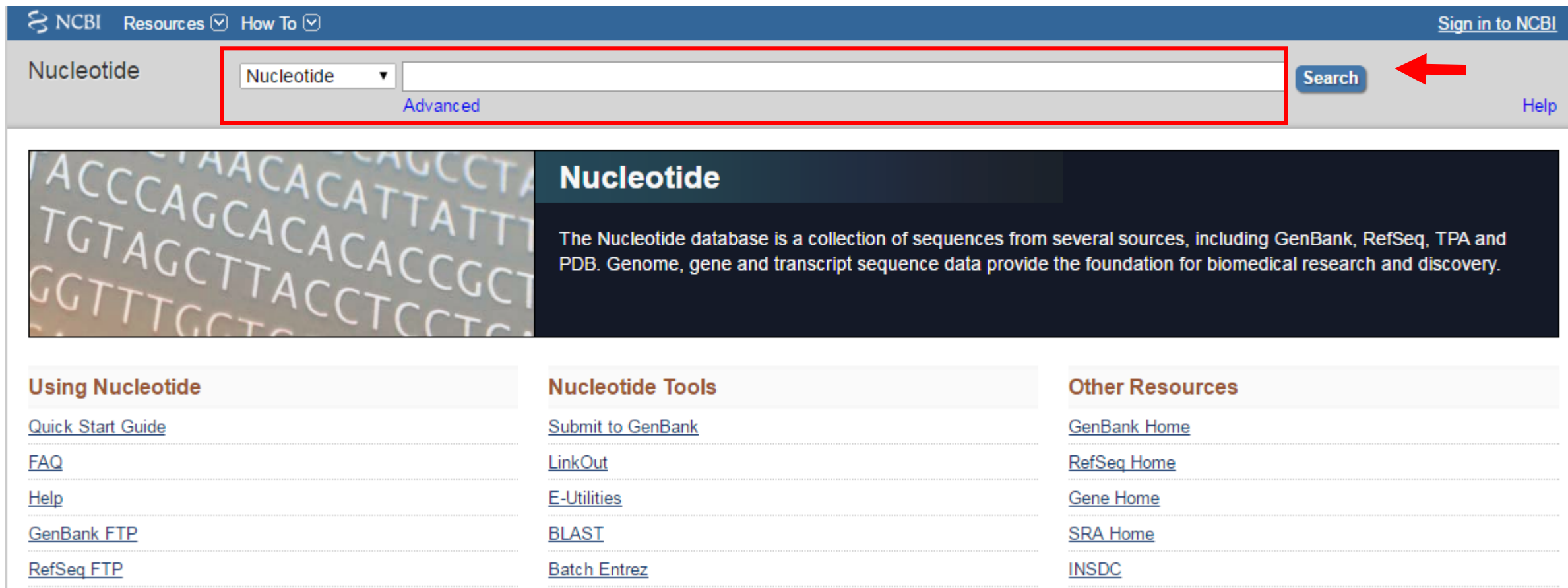
如何检索

- 1) Entrez Nucleotide
- 2) Blast
- 3) NCBI e-utilities
- 4) FTP: <ftp://ftp.ncbi.nlm.nih.gov/ncbi-asn1>
ftp://ftp.ncbi.nlm.nih.gov/genbank

如何检索 (Entrez Nucleotide)

<http://www.ncbi.nlm.nih.gov/nuccore>

Entrez是NCBI的为用户提供整合的访问序列，定位，分类，和结构数据的搜索和检索系统



NCBI Resources How To Sign in to NCBI

Nucleotide Nucleotide Search Help

Nucleotide

The Nucleotide database is a collection of sequences from several sources, including GenBank, RefSeq, TPA and PDB. Genome, gene and transcript sequence data provide the foundation for biomedical research and discovery.

Using Nucleotide

- [Quick Start Guide](#)
- [FAQ](#)
- [Help](#)
- [GenBank FTP](#)
- [RefSeq FTP](#)

Nucleotide Tools

- [Submit to GenBank](#)
- [LinkOut](#)
- [E-Utilities](#)
- [BLAST](#)
- [Batch Entrez](#)

Other Resources

- [GenBank Home](#)
- [RefSeq Home](#)
- [Gene Home](#)
- [SRA Home](#)
- [INSDC](#)

Entrez

- 用以整合NCBI数据库中信息的搜寻和检索工具
- 同时也提供序列和染色体图谱的图形视图。
- 这些数据库包括核酸序列，蛋白序列，大分子结构，全基因组，和通过PubMed检索的MEDLINE。
- Entrez的一个强大和独特的特点是检索相关的序列，结构，和参考文献的能力。
- 参考文献通过PubMed获得

基本检索功能

1. 限定词检索
2. 特殊标识符检索 (ACCESSION, GI)
3. 特定短语
4. 范围检索
5. 高级检索

Tips: 其余数据库的检索基本通用。

1st: 限定词检索

- 1) protein name : insulin
- 2) gene name, or gene symbol : p53,CREB1
- 3) Species name: homo sapien
- 4) a submitter or author name in the following format will produce the best results.
 - Smith JR (last name followed by initials, no punctuation)

Nucleotide

Nucleotide

p53

✕

Search

[Create alert](#) [Advanced](#)[Help](#)

NCBI is phasing out sequence GI numbers in September 2016. Please use accession.version! [Read more...](#)

Species

Animals (28,646)
Plants (408)
Fungi (971)
Protists (323)
Bacteria (43,451)
Archaea (12)
Viruses (334)
[Customize ...](#)

Molecule types

genomic DNA/RNA (62,418)
mRNA (16,111)
rRNA (2)
[Customize ...](#)

Source databases

INSDC (GenBank) (36,155)
RefSeq (44,920)
[Customize ...](#)

Genetic

Summary ▾ 20 per page ▾ Sort by Default order ▾

Send to: ▾

Filters: [Manage Filters](#)

See **p53** CG33336 gene product from transcript CG33336-RB in the Gene database
p53 reference sequences [Transcript \(4\)](#) [Protein \(4\)](#)

Items: 1 to 20 of 81265

<< First < Prev Page 1 of 4064 Next > Last >>

Found 82105 nucleotide sequences. Nucleotide (81265) EST ([804](#)) GSS ([36](#))

☐ [Mus musculus mutant p53 mRNA, complete cds](#)

1. 1,429 bp linear mRNA

Accession: AB021961.1 GI: 5421849

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Mus musculus mRNA for p53, complete cds](#)

2. 1,457 bp linear mRNA

Accession: AB020317.1 GI: 5421827

[GenBank](#) [FASTA](#) [Graphics](#)

Results by taxon

Top Organisms [\[Tree\]](#)

Mycobacterium tuberculosis (8430)
Homo sapiens (5723)
Pseudomonas aeruginosa (5384)
synthetic construct (4058)
Acinetobacter baumannii (3858)
All other taxa (53812)

[More...](#)

Find related data

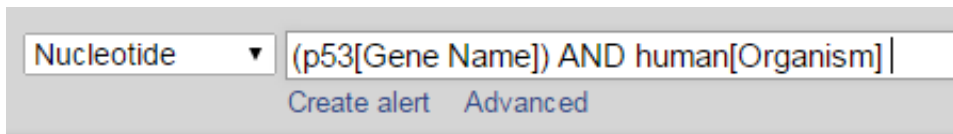
Database: [Select](#)

[Find items](#)

[Search details](#)

检索限定词

- 1、基因名称的检索限定词：[GENE] or [GENE NAME]
- 2、生物体名称的检索限定词：[ORGN] or [ORGANISM]
- 3、作者姓名的检索限定词：[AUTH] or [AUTHOR]



Nucleotide ▼ (p53[Gene Name]) AND human[Organism] |
Create alert Advanced

格式：gene name[GENE] AND human[ORGN]

逻辑操作符

- 与： AND
 - contactin AND neurofascin
- 或： OR
 - contactin OR neurofascin
- 非： NOT
 - contactin NOT neurofascin

不同的检索词之间可以通过逻辑操作符连接。

2nd: 标识符检索

- 1. 基因信息号 (GI)

为一串数字，如6995995

- 2. Accession number:

a) GenBank/EMBL/DDBJ序列接收号

1个字母 + 5个数字 如 U12345

2个字母 + 6个数字 如AY12345

b) PDB序列接受号

1个数字 + 3个字母 如1TUP

☐ [Homo sapiens tumor protein p53 \(TP53\), RefSeqGene \(LRG_321\) on chromosome 17](#)

1. 32,772 bp linear DNA

Accession: NG_017013.2 GI: 383209646

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 1, mRNA](#)

2. 2,591 bp linear mRNA

Accession: NM_000546.5 GI: 371502114

[GenBank](#) [FASTA](#) [Graphics](#)

标识符检索

- 2) Accession number

- c) RefSeq 序列号

- mRNA记录 (NM_*)
 - Contig or scaffold, clone-based or WGS (NT_*)
 - 完整基因组或染色体 (NC_*)
 - 基因组局部区域 (NG_*)
 - 人类基因组序列注释加工得到的序列模型 (XM, XP, XR_*)

`/transcript_id="NM_001143990.1"`



RefSeq: NCBI Reference Sequence Database

A comprehensive, integrated, non-redundant, well-annotated set of reference sequences including genomic, transcript, and protein.

Table 1. [RefSeq](#) accession numbers and molecule types.

Accession prefix	Molecule type	Comment
AC_	Genomic	Complete genomic molecule, usually alternate assembly
NC_	Genomic	Complete genomic molecule, usually reference assembly
NG_	Genomic	Incomplete genomic region
NT_	Genomic	Contig or scaffold, clone-based or WGS ^a
NW_	Genomic	Contig or scaffold, primarily WGS ^a
NS_	Genomic	Environmental sequence
NZ_ ^b	Genomic	Unfinished WGS
NM_	mRNA	
NR_	RNA	
XM_ ^c	mRNA	Predicted model
XR_ ^c	RNA	Predicted model
AP_	Protein	Annotated on AC_ alternate assembly
NP_	Protein	Associated with an NM_ or NC_ accession
YP_ ^c	Protein	
XP_ ^c	Protein	Predicted model, associated with an XM_ accession
ZP_ ^c	Protein	Predicted model, annotated on NZ_ genomic records

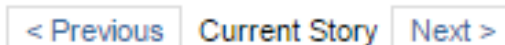
^a Whole Genome Shotgun sequence data.

^b An ordered collection of [WGS sequence](#) for a genome.

^c Computed.

<https://www.ncbi.nlm.nih.gov/genbank/samplerecord/>

须知



NCBI is phasing out sequence GIs - use Accession.Version instead!

Wednesday, March 2, 2016

As of September 2016, the integer sequence identifiers known as "GIs" will no longer be included in the GenBank, GenPept, and FASTA formats supported by NCBI for sequence records. The FASTA header will be further simplified to report only the sequence accession.version and record title for accessions managed by the International Sequence Database Collaboration (INSDC) and NCBI's Reference Sequence (RefSeq) project. As NCBI makes this transition, we encourage any users who have workflows that depend on GI's to begin planning to use accession.version identifiers instead. After September 2016, any processes solely dependent on GIs will no longer function as expected.

3rd: 范围检索

1. 序列接收号范围检索:

- AF114696:AF114714[ACCN]
- 序列接收号的检索限定词为[ACCN]或者[ACCESSION]

2. 序列长度范围检索

- 3000:4000[SLEN]

3. 日期范围检索

1. 2012/01: 2015/12/31[MDAT] or [PDAT]

[ACCN] Search Tip : The letters in the accession number can be written in upper- or lowercase. RefSeq accessions must contain an underscore bar between the letters and the numbers, e.g., NM_002111.

GSS

GSS

AF114696:AF114714[ACCN]



Search

Create alert Limits Advanced

Help

Summary ▾ 20 per page ▾ Sort by Default order ▾

Send to: ▾

Filter your results:

All (19)

Bacteria (0)

mRNA (0)

[Manage Filters](#)

Analyze these sequences

Run BLAST

Find related data

Database: Select ▾

Find items

Items: 19

☐ [AF114709 Felis catus Fca215 Felis catus genomic clone AMELX genomic survey sequence](#)

1. 356 bp linear DNA

Accession: AF114709.2 GI: 7478248

[GSS](#) [GenBank](#) [FASTA](#)

☐ [AF114710 Felis catus Fca215 Felis catus genomic clone RPS4X genomic survey sequence](#)

2. 419 bp linear DNA

Accession: AF114710.2 GI: 7477749

[GSS](#) [GenBank](#) [FASTA](#)

☐ [AF114714 Felis catus Fca215 Felis catus genomic clone SMCY genomic survey sequence](#)

3. 292 bp linear DNA

Accession: AF114714.1 GI: 5106975

[GSS](#) [GenBank](#) [FASTA](#)

检索限定词还是很重要的

Nucleotide ▾

AF114696:AF114714



Search

Create alert Advanced

sequence GI numbers in September 2016. Please use accession.version! [Read more...](#)



The following terms were not found in Nucleotide: AF114696, AF114714.



No items found.

Search details

(AF114696[All Fields]
Fields])

4th:高级检索



Nucleotide

Nucleotide

Advanced

Search

Help



Nucleotide

The Nucleotide database is a collection of sequences from several sources, including GenBank, RefSeq, TPA and PDB. Genome, gene and transcript sequence data provide the foundation for biomedical research and discovery.

Using Nucleotide

- [Quick Start Guide](#)
- [FAQ](#)
- [Help](#)
- [GenBank FTP](#)
- [RefSeq FTP](#)

Nucleotide Tools

- [Submit to GenBank](#)
- [LinkOut](#)
- [E-Utilities](#)
- [BLAST](#)
- [Batch Entrez](#)

Other Resources

- [GenBank Home](#)
- [RefSeq Home](#)
- [Gene Home](#)
- [SRA Home](#)
- [INSDC](#)

GenBank数据库高级检索功能



[NCBI](#) [Resources](#) [How To](#) [Sign in to NCBI](#)

[Nucleotide Home](#) [Help](#)

Nucleotide Advanced Search Builder

Use the builder below to create your search

[Edit](#) [Clear](#)

Builder

All Fields

hepatitis b

-

[Show index list](#)

AND

All Fields

- +

[Show index list](#)

Search

 or [Add to history](#)

History

[Download history](#) [Clear history](#)

Search	Add to builder	Query	Items found	Time
#7	Add	Search "ras" Field: Gene Name	501	19:55:04
#4	Add	Search "ras"[gene]	501	19:53:47
#1	Add	Search "ras"	109161	19:53:09

Nucleotide Advanced Search Builder

Use the builder below to create your search

[Edit](#)

[Clear](#)

Builder

All Fields



AND

All Fields



Search

or [Add to history](#)



- Accession
- All Fields
- Assembly
- Author
- BioProject
- BioSample
- Breed
- Component Accession
- Cultivar
- Division
- EC/RN Number
- Feature key
- Filter
- Gene Name
- Isolate
- Issue
- Journal
- Keyword
- Modification Date
- Organism

- Page Number
- Primary Accession
- Primary Organism
- Properties
- Protein Name
- Publication Date
- SeqID String
- Sequence Length
- Strain
- Substance Name
- Text Word
- Title
- Volume

GenBank数据库高级检索功能



乙型肝炎索引

hepatitis b [Edit](#) [Clear](#)

Builder

All Fields ▼ hepatitis b [Hide index list](#)

hepatitis b (166003)
hepatitis b/hiv co infected patient (4)
hepatitis b and c (642)
hepatitis b and c virus (271)
hepatitis b and c viruses (320)
hepatitis b antigen (1)
hepatitis b associated (1)
hepatitis b associated hepatocellular (1)
hepatitis b associated hepatocellular carcinoma (1)
hepatitis b carrier (1)

[Previous 200](#)
[Next 200](#)
[Refresh index](#)

AND ▼ All Fields ▼ [Show index list](#)

[Search](#) or [Add to history](#)

如何过滤搜索结果



Species

Animals (54,306)

Plants (2)

Fungi (52)

Protists (1)

Bacteria (12)

Viruses (88,421)

Customize ...

Molecule types

genomic DNA/RNA (95,334)

mRNA (54,205)

Customize ...

Source databases

INSDC (GenBank) (162,591)

RefSeq (3,401)

Customize ...

Genetic compartments

Mitochondrion (11)

Plasmid (2)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

[Clear all](#)

[Show additional filters](#)

Summary ▾ 20 per page ▾ Sort by Default order ▾ Send to: ▾

See [b black](#) in the Gene database
b reference sequences [Transcript \(3\)](#) [Protein \(3\)](#)

Items: 1 to 20 of 166003

<< First < Prev Page 1 of 8301 Next > Last >>

Found 166781 nucleotide sequences. Nucleotide (166003) EST (283) GSS (495)

☐ [Hepatitis B virus subtype ADR clone TL29-A3 truncated large envelope protein gene, complete cds](#)

1. 1,404 bp linear DNA

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [WO 2015133613-A/5: Hepatitis B virus like particle and method for a screening of drug for treating hepatitis B using the same](#)

2. 4,000 bp linear DNA

Accession: HZ376673.1 GI: 970255163

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [JP 2014140367-A/38: Primer for amplifying hepatitis B virus DNA and method for detecting hepatitis B virus by use thereof](#)

3. 3,215 bp linear DNA

Accession: HW610289.1 GI: 726965150

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [JP 2014140367-A/37: Primer for amplifying hepatitis B virus DNA and method for detecting hepatitis B virus by use thereof](#)

4. 3,215 bp linear DNA

Accession: HW610288.1 GI: 726965149

[GenBank](#) [FASTA](#) [Graphics](#)

Facets/Filters

如何过滤搜索结果 (Organism)



Species Summary ▾ 20 per page ▾ Sort by Default order ▾

Animals (18,104,022)
Plants (6,528,053)
Fungi (8,339,996)
Protists (3,215,785)
Bacteria (230,345,242)
Archaea (2,522,481)
Viruses (3,453,766)
Customize ...

Items: 1 to 20 of 283622511

<< First < Prev Page 1

☐ [PREDICTED: semaphorin-3D-like, partial \[Sinocyclocheilus grahami\]](#)

1. 230 aa protein

Species

- ☒ Animals
- ☒ Plants
- ☒ Fungi
- ☒ Protists
- ☒ Bacteria
- ☒ Archaea
- ☒ Viruses

Add

Mus musculus

Mus musculus

Mus musculus domesticus

Mus musculus praetextus

Mus musculus castaneus

Mus musculus hortulanus

Mus musculus musculus

Mus musculus spretus

See all results

1020615144
[STA](#) [Graphics](#)
[associated activator of morphogenesis 2-like, p](#)

1020615139
[STA](#) [Graphics](#)
[peat-containing protein 30-like, partial \[Sinoc](#)

1020615134
[STA](#) [Graphics](#)
[subunit pICln-like, partial \[Sinocyclocheilus g](#)

1020615128
[FASTA](#) [Graphics](#)
[3-like, partial \[Sinocyclocheilus grahami\]](#)

GI: 1020615123
[FASTA](#) [Graphics](#)

Species Summary ▾ 20 per page ▾ Sort by Default order ▾

Animals (18,104,022)
Plants (6,528,053)
Fungi (8,339,996)
Protists (3,215,785)
Bacteria (230,345,242)
Archaea (2,522,481)
Viruses (3,453,766)
Customize ...

Items: 1 to 20 of 283622511

<< First < Prev Page 1

☐ [PREDICTED: semaphorin-3D-like, partial \[Sinocyclocheilus grahami\]](#)

1. 230 aa protein

Species

- ☒ Animals
- ☒ Plants
- ☒ Fungi
- ☒ Protists
- ☒ Bacteria
- ☒ Archaea
- ☒ Viruses
- ☒ Mus musculus

Add

Show

1020615144
[STA](#) [Graphics](#)
[associated activator of morphogenesis 2-like, p](#)

1020615139
[STA](#) [Graphics](#)
[peat-containing protein 30-like, partial \[Sinoc](#)

1020615134
[STA](#) [Graphics](#)
[subunit pICln-like, partial \[Sinocyclocheilus g](#)

1020615128
[STA](#) [Graphics](#)

Species

clear

Summary ▾ 20 per page ▾ Sort by Default order ▾

Send to: ▾

Animals (312,670)

Plants (0)

Fungi (0)

Protists (0)

Bacteria (0)

Archaea (0)

Viruses (0)

✓ Mus musculus (312,671)

Customize ...

Source databases

PDB (11,170)

RefSeq (78,319)

UniProtKB / Swiss-Prot (16,781)

Customize ...

Genetic compartments

Mitochondrion (2,863)

Plasmid (24)

Sequence length

Custom range...

Molecular weight

Custom range...

Release date

Custom range...

Revision date

Custom range...

Clear all

Show additional filters

Items: 1 to 20 of 312671

<< First < Prev Page 1 of 15634 Next > Last >>

Filters activated: Mus musculus. [Clear all](#)

☐

[lipase_member O4 precursor \[Mus musculus\]](#)

1. 398 aa protein

Accession: NP_001310315.1 GI: 1020284562

[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐

[integrin alpha-5 isoform 1 preproprotein \[Mus musculus\]](#)

2. 1053 aa protein

Accession: NP_034707.5 GI: 1020159030

[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐

[Gm8978 protein precursor \[Mus musculus\]](#)

3. 399 aa protein

Accession: NP_001310180.1 GI: 1019286538

[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐

[lipase_member O1 precursor \[Mus musculus\]](#)

4. 399 aa protein

Accession: NP_001309994.1 GI: 1018976630

[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐

[RecName: Full=DENN domain-containing protein 1B; AltName: Full=Connecdenn 2](#)

5. 766 aa protein

Accession: Q3U1T9.3 GI: 1018807853

[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐

[RecName: Full=Selenoprotein N; Short=SelN; Flags: Precursor](#)

6. 557 aa protein

Accession: D3Z2R5.2 GI: 1018740189

[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

解螺旋

如何过滤搜索结果 (Molecule type)



Molecule types

genomic

DNA/RNA (107,209,960)

mRNA (38,091,526)

rRNA (176,537)

Customize ...

如何过滤搜索结果 (Source database)



Source databases
INSDC (GenBank) (168,713,572)
RefSeq (32,892,401)
Customize ...

Accession: KV408990.1 GI: 1020925611
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Scleropages formosus breed green](#)

Genetic compartment
Chloroplast (86)
Mitochondrion
Plasmid (150,1)
Plastid (946,74)

Sequence length
Custom range.

Release date
Custom range.

Revision date
Custom range.

[Clear all](#)

Source databases

☐ DDBJ
☐ EMBL
☐ GenBank
☒ INSDC (GenBank)
☐ PDB
☐ PIR
☒ RefSeq
☐ UniProtKB / Swiss-Prot

Show

Nucleotide

Source databases
PDB (321,441)
RefSeq (63,304,148)
UniProtKB / Swiss-Prot (551,013)
Customize ...

☐ [PREDICTED: semaphorin-3D-like, partial](#)

1. **230 aa protein**
Accession: XP_016120447.1 GI: 1020615144
[FASTA](#) [Graphics](#)

Genetic compartment
Chloroplast (69)
Mitochondrion
Plasmid (840,4)
Plastid (807,67)

Sequence length
Custom range.

Molecular weight
Custom range.

Release date
Custom range.

Source databases

☒ PDB
☒ RefSeq
☒ UniProtKB / Swiss-Prot
☐ DDBJ
☐ EMBL
☐ GenBank
☐ PIR

Show

Protein

正常检索结果



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Nucleotide Nucleotide human p53 Search Save search Advanced Help

Species
Animals (11,507)
Plants (154)
Fungi (470)
Protists (121)
Bacteria (13,210)
Archaea (14)
Viruses (240)
Customize ...

Molecule types
genomic DNA/RNA (22,233)
mRNA (3,527)
rRNA (2)
Customize ...

Source databases
GenBank (9,089)
RefSeq (14,126)
Customize ...

Genetic compartments
Mitochondrion (2)
Plasmid (24)

Sequence length
Custom range...

Release date

Display Settings: Summary, 20 per page, Sorted by Default order Send to: Filters: Manage Filters

Results: 1 to 20 of 26102 << First < Prev Page 1 of 1306 Next > Last >>

Found 26333 nucleotide sequences. Nucleotide (26102) EST (196) GSS (35)

☐ [Homo sapiens mRNA for P53, complete cds](#)

1. 2,451 bp linear mRNA
Accession: AB082923.1 GI: 23491728
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens p53 \(p53\) gene, exon 7 and partial cds](#)

2. 110 bp linear DNA
Accession: JF923572.1 GI: 349734069
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens p53 \(p53\) gene, exon 6 and partial cds](#)

3. 113 bp linear DNA
Accession: JF923571.1 GI: 349734067
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens p53 \(p53\) gene, exon 5 and partial cds](#)

4. 183 bp linear DNA
Accession: JF923570.1 GI: 349734065
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens p53 \(p53\) gene, exon 4 and partial cds](#)

Results by taxon
Top Organisms [Tree](#)
Homo sapiens (5523)
Pseudomonas aeruginosa (1823)
Mycobacterium tuberculosis (884)
Klebsiella pneumoniae (868)
Acinetobacter baumannii (829)
All other taxa (16175)
More...

Find related data
Database: Select
Find items

Search details
("Homo sapiens"[Organism] OR human[All Fields]) AND p53[All Fields]
Search See more

如何调整搜索结果的显示方式



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Nucleotide Nucleotide human p53 Search Save search Advanced Help

Species: Animals (11,507), Plants (154), Fungi (470), Protists (121), Bacteria (13,210), Archaea (14), Viruses (240), Customize ...

Molecule types: genomic DNA/RNA (22,233), mRNA (3,527), rRNA (2), Customize ...

Source databases: GenBank (9,089), RefSeq (14,126), Customize ...

Genetic compartments: Mitochondrion (2), Plasmid (24)

Sequence length: Custom range...

Release date

Display Settings: Summary, 20 per page, Sorted by Default order

Format: Summary (selected), GenBank, GenBank (full), FASTA, FASTA (text), ASN.1, Revision History, Accession List, GI List

Items per page: 5, 10, 20 (selected), 50, 100, 200

Sort by: Default order (selected), Accession, Date Modified, Date Released, Organism Name, Taxonomy ID

Send to: Filters: Manage Filters

Results by taxon: Top Organisms [Tree]
Homo sapiens (5523)
Pseudomonas aeruginosa (1823)
Mycobacterium tuberculosis (884)
Klebsiella pneumoniae (868)
Acinetobacter baumannii (829)
All other taxa (16175)
More...

Find related data: Database: Select Find items

Search details: ("Homo sapiens"[Organism] OR human[All Fields]) AND p53[All Fields]

Accession: JF923572.1 GI: 349734069
GenBank FASTA Graphics

3. 113 bp linear DNA
Accession: JF923571.1 GI: 349734067
GenBank FASTA Graphics

4. 183 bp linear DNA
Accession: JF923570.1 GI: 349734065
GenBank FASTA Graphics

5. 183 bp linear DNA
Accession: JF923569.1 GI: 349734064
GenBank FASTA Graphics

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☐ Collections

Download 207 items.

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- 1. NCBI及GenBank的介绍
- 2. 如何检索
- 3. 解释检索结果
- 4. 下载选定结果
- 4. NCBI其他资源介绍

基因的详细信息

默认显示Genbank格式



[Display Settings:](#) ☒ GenBank [Send:](#) ☐

Homo sapiens mRNA for P53, complete cds

GenBank: AB082923.1

[FASTA](#) [Graphics](#)

[Go to:](#) ☐

LOCUS AB082923 2451 bp mRNA linear PRI 01-APR-2003

DEFINITION Homo sapiens mRNA for P53, complete cds.

ACCESSION AB082923

VERSION AB082923.1 GI:23491728

KEYWORDS .

SOURCE Homo sapiens (human)

ORGANISM [Homo sapiens](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
Catarrhini; Hominidae; Homo.

REFERENCE 1

AUTHORS Azuma, K., Shichijo, S., Maeda, Y., Nakatsura, T., Nonaka, Y., Fujii, T.,
Koike, K. and Itoh, K.

TITLE Mutated p53 gene encodes a nonmutated epitope recognized by
HLA-B*4601-restricted and tumor cell-reactive CTLs at tumor site

JOURNAL Cancer Res. 63 (4), 854-858 (2003)

PUBMED [12591737](#)

REFERENCE 2 (bases 1 to 2451)

AUTHORS Shichijo, S. and Itoh, K.

TITLE Direct Submission

JOURNAL Submitted (26-MAR-2002) Shigeki Shichijo, Kurume Univ. School of
Med., Dep. Immunol.; 67-Asahi-machi, Kurume, Fukuoka 830-0011,
Japan (E-mail: shichijo@med.kurume-u.ac.jp, Tel: 81-942-31-7551,
Fax: 81-942-31-7699)

FEATURES Location/Qualifiers

source 1..2451

Change region shown

Customize view

Analyze this sequence

[Run BLAST](#)

[Pick Primers](#)

[Highlight Sequence Features](#)

[Find in this Sequence](#)

Articles about the TP53 gene

[\[Association of polymorphic markers Arg72Pro of TP53 and T309g of MDM2 \[Mol Biol \(Mosk\). 2014\]](#)

[Ferroptosis as a p53-mediated activity during tumour suppression. \[Nature. 2015\]](#)

[Clinicopathological importance of Ki-67, p27, and p53 expression in gastric. \[Turk J Med Sci. 2015\]](#)

[See all...](#)

Pathways for the TP53 gene

[TP53 Regulates Metabolic Genes](#)

[Transcriptional Regulation by TP53](#)

[Sphingolipid signaling pathway](#)



可人工调整显示方式

GenBank ▾

Send: ▾

Homo sapiens mRNA for P53, complete cds

GenBank: AB082923.1

[FASTA](#) [Graphics](#)

[Go to:](#) ▾

LOCUS AB082923 2451 bp mRNA linear PRI 01-APR-2003
DEFINITION Homo sapiens mRNA for P53, complete cds.
ACCESSION AB082923
VERSION AB082923.1 GI:23491728
KEYWORDS .
SOURCE Homo sapiens (human)
ORGANISM [Homo sapiens](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
Catarrhini; Hominidae; Homo.
REFERENCE 1
AUTHORS Azuma,K., Shichijo,S., Maeda,Y., Nakatsura,T., Nonaka,Y., Fujii,T.,
Koike,K. and Itoh,K.
TITLE Mutated p53 gene encodes a nonmutated epitope recognized by
HLA-B*4601-restricted and tumor cell-reactive CTLs at tumor site
JOURNAL J. Biol. Chem. 273(14): 8514-8520 (1998)

Change region shown

☒ Whole sequence

☐ Selected region

from: to:

Update View

Customize view

Basic Features

☐ Default features

☒ Gene, RNA, and CDS features only

Features added by NCBI

☐ 24 SNPs

☐ 3 conserved domains

Display options

☒ Show sequence

☐ Show reverse complement

Update View

完整的基因记录 (Gene Report)

- 基因位点 (Locus)
- 基因定义 (Definition)
- 基因存取号 (Accession)
- 核酸编号(NID)
- 关键词 (Keywords)
- 来源 (Source)
- 物种分类 (Organism)
- 参考文献 (Reference)
- 著者 (Author)
- 题目 (Title)
- 期刊 (Journal)
- Medline存取号 (Medline)
- 序列特征 (Features)
- 基因 (Gene)
- CDS (cDNA)
- 等位基因 (Allele)
- 对等的肽 (Mat-Peptide)
- 碱基数 (Base Count)
- 原序列 (Origin)

特性表

FEATURES

source

Location/Qualifiers

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 /db_xref="taxon:9606"
 /clone="7F"
 /cell_line="SW620"
 /cell_type="colon cancer"

gene

1..2451

/gene="p53"

CDS

64..1245

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/codon_start=1

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/db_xref="GI:23491729"

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 AIYKQSQHMTVVRRCPHHERCSDSDGLAPPQHLIRVEGNLRVEYLDDRNTFRHSVVV
 PYEPPEVGSDCTTIHYNMCMNSMGGMNRRLPILTIITLEDSSGNLLGRNSFEVHVCA
 CPGRDRRTEENLRKKGEPHHELPPGSTKRALSNNNTSSSPQPKKKPLDGEYFTLQIRG
 RERFEMFRELNEALELKDAQAGKEPGGSRAHSSHLKSKKGQSTSRHKKLMFKTEGPD
 S"

GenBank记录中特性表中的主要关键词:



关键词	解 释	关键词	解 释
misc_feature	生物学特性无法用特性表关键词描述的序列	promoter	转录起始区
misc_difference	序列特性无法用特性表关键词描述的序列	CAAT_signal	真核启动子上游的CAAT盒,与RNA结合相关
conflict	同一序列在不同的研究中有在位点或区域上有差异	TATA_signal	真核启动子的TATA盒
unsure	序列不能确定的区域	-35_signal	原核启动子中的-35框
old_sequence	该序列对以前的版本做过修订	-10_signal	原核启动子的Pribow盒
variation	包含稳定突变的序列	GC_signal	真核启动子的GC盒
modified_base	修饰过的核苷酸	RBS	核糖体结合位点
gene	已识别为基因或已命名的序列区域	polyA_signal	RNA转录本的剪切识别位点
misc_signal	无法用信号特性关键词描述的信号序列	enhancer	增强子

关键词	解 释	关键词	解 释
attenuator	与转录终止有关的序列	CDS	蛋白质编码序列
terminator	转录终止序列	sig_peptide	编码信号肽的序列
rep_origin	双链DNA复制起始区	transit_peptide	转运蛋白编码序列
misc_RNA	无法用RNA关键词描述的转录物或RNA产物	mat_peptide	编码成熟肽的序列
prim_transcript	初始转录本	intron	内含子
precursor_RNA	前体RNA	polyA_site	RNA转录本的多聚腺苷酸化位点
mRNA	信使RNA	rRNA	核糖体RNA
5' clip	前体转录本中被剪切掉的5'端序列	tRNA	转运RNA
3' clip	前体转录本中被剪切掉的3'端序列	scRNA	小细胞质RNA
5'UTR	5'非翻译区	snRNA	小核RNA
3'UTR	3'非翻译区	snoRNA	加工和修饰rRNA的小核RNA
exon	外显子		



解螺旋

关键词	解 释	关键词	解 释
immunoglobulin_related		repeat_unit	单个的重复元件
C_region	免疫相关蛋白上的不变区	LTR	长末端重复序列
D_segment	免疫球蛋白重链的可变区, T细胞受体β链	Satellite	卫星重复序列
J_segment	免疫球蛋白重链、轻链以及 T细胞α、β、γ的结合链	misc_binding	无法描述的核酸序列结合位点
N_region	插入重排免疫球蛋白片段间的核苷酸	primer_bind	复制、转录的引物结合位点
S_region	免疫球蛋白重链的开关区	protein_bind	蛋白质结合区
V_region	编码免疫球蛋白的可变区N末端的序列	STS	测序标签位点
V_segment	编码免疫球蛋白的可变区的序列		
		misc_recomb	无法用重组特性关键词描述的重组事件
repeat_region	基因组中所包含的重复序列	iDNA	通过重组所消除的DNA



关键词	解 释	关键词	解 释
misc_structure	无法用结构关键词描述的核酸序列高级结构或构型		
stem_loop	发夹结构		
D_loop	线粒体中DNA中的取代环		



GenBank记录中特性表中的限定词:

限定词	含 义	限定词	含 义
/allele=	给定基因的等位基因	/codon_start=	相对于序列第一个碱基，编码序列密码子的偏移量
/bound_moiety=	嵌合范围	/country=	DNA样本的来源国
/cell_type=	获得序列的细胞类型	/db_xref=	其他数据库信息的交叉索引号
/citation=	已被引用的参考文献数	/direction=	DNA复制方向
/clone_lib=	获得序列的克隆文库	/environmental_sample=	序列直接从环境材料中获得而没有指明来源物种

限定词	含 义	限定词	含 义
/exception=	指明DNA序列未按通常的生物学规律翻译，如RNA编辑	/PCR_conditions=	描述PCR的反应条件
/frequency=	在种群中发生变异的频率	/pop_variant=	获得序列的群体变异种名称
/germline	如果序列是DNA并来源于免疫球蛋白家族，则表示该序列来源于未重排DNA	/product=	序列编码产物的名称
/insertion_seq=	序列来源于某种插入元件	/anticodon=	tRNA反义密码子的位置及它所编码的氨基酸
/isolate=	序列来源的生物个体	/cell_line=	获得序列的细胞系
/lab_host=	为扩增序列来源物种所用的实验室宿主	/chromosome=	获得序列的染色体
/macronuclear	指明DNA来源于染色体分化的大核期	/clone=	获得序列的克隆子
/note=	评论及附加信息	/codon=	指出与参考密码子不同的密码子
/organelle=	获得序列的细胞器	/EC_number=	序列产物的酶学编号



解螺旋

限定词	含 义	限定词	含 义
/cons_splice=	区分内含子剪切位点和“5'-GT.AG-3'”剪切位点	/map=	相关特性在基因图谱上的位置
/cultivar=	所获序列植物的栽培变种	/mod_base=	被修饰碱基的简写
/dev_stage=	序列来源于某种生物的特定发育阶段	/number=	从5'→3'注明遗传元件的顺序
/evidence=	序列特性来源于实验还是推理	/organism=	提供测序用遗传物质的物种的科学名称
/focus	指出在记录中的来源特性在其他物种中还有不同的来源特性	/phenotype=	序列特性所导致的表型
/function=	序列所代表的功能	/plasmid=	获得序列的质粒名称
/haplotype=	序列来源于某种物种的单倍体	/protein_id=	蛋白质的检索号
/isolation_source=	描述序列来源物种的生理、环境和地理信息	/proviral	整合在基因组中的前病毒
/label=	序列特性的俗名	/rearranged	如果序列是DNA并来源于免疫球蛋白家族,则表示该序列来源于重排DNA

解螺旋

限定词	含 义	限定词	含 义
/rpt_family=	重复序列	/transposon=	转座子
/rpt_unit=	指明重复区域的重复元件构成	/variety=	获得序列的生物变种
/serotype=	同一物种的不同血清学特征	/pseudo	假基因
/sex=	获得序列的物种性别	/replace=	表明特性间的间隔序列已被替换
/specimen_voucher=	指明来源物种保存于什么地方	/rpt_type=	重复序列的组织方式
/strain=	获得序列的菌株	/sequenced_mol=	获得序列的分子类型
/sub_species=	获得序列的来源物种的亚种	/serovar=	同一原核生物的血清学特征
/tissue_lib=	获得序列组织库	/specific_host=	获得序列的天然宿主
/transgenic	指明物种的来源特性是否是转基因受体	/standard-name=	特性的通用名称
/transl_except=	标明序列中未按指定密码子表翻译的氨基酸的位置	/sub_clone=	获得序列的亚克隆



解螺旋

限定词	含 义	限定词	含
/sub_strain=	获得序列的来源微生物亚种		
/tissue_type=	获得序列组织类型		
/translation=	按通用或指定的密码子表翻译的氨基酸序列		
/transl_table=	描述在翻译中与通用密码表不同的密码表		
/usedin=	表明该特性在其他检索中也被使用		
/virion	病毒颗粒		



解螺旋

显示FASTA格式

Homo sapiens mRNA for P53, complete cds

GenBank: AB082923.1

[FASTA](#) [Graphics](#)

系统

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Nucleotide Nucleotide Search

Advanced Help

Display Settings: FASTA

Send:

Homo sapiens mRNA for P53, complete cds

GenBank: AB082923.1

[GenBank](#) [Graphics](#)

```
>gi|23491728|dbj|AB082923.1| Homo sapiens mRNA for P53, complete cds
CGTGCTTTCCACGACGGTGACACGCTTCCCTGGATTGGCCAGACTGCCTTCCGGGTCACTGCCATGGAGG
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TCTGAAAAACAACGTTCTGTCCCTTTGCGTCCCAAGCAATGGATGATTGATGCTGTCCCGGACGAT
ATTGAACAATGGTTCACTGAAGACCCAGGTCCAGATGAAGCTCCCAGAATGCCAGAGGCTGCTCCCGCG
TGGCCCTGCAACAGCAGCTCTACACCGCGGCCCCCTGCAACAGCCCCCTCTGGCCCTGTCTCTTC
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GACATAGTGTGGTGGTGCCCTATGAGCGCCTGAGGTTGGCTCTGACTGTACCACCATCCACTACAACCTA
CATGTGTAACAGTTCCCTGCATGGGCGGCATGAACCGGAGGCCCATCTCACCATCATCACACTGGAAGAC
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GCACAGAGGAAGAGAATCTCCGCAAGAAAGGGGAGCCTCACACGAGCTGCCCCAGGGAGCACTAAGCG
AGCACTGTCCAACAACACCACTCTCTCCCCAGCCAAAGAAAGAAACCACTGGATGGAGAATATTTTACC
CTTCAGATCCGTGGGCGTGAGCGCTTCGAGATGTTCCGAGAGCTGAATGAGGCTTGGAACTCAAGGATG
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TACCTCCCGCCATAAAAACTCATGTTCAAGACAGAAAGGGCTGACTCAGACTGACATTCTCCACTTCTT
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CTTGCTTGCAATAGGTGTGCGTCAGAAGCACCCAGGACTTCCATTTGCTTTGTCCCGGGGCTCCACTGAA
CAAGTTGGCTGCACTGGTGTGTTGTTGTTGGGAGGAGGATGGGGAGTAGGACATACCAGCTTAGATTTT
AAGGTTTTTACTGTGAGGATGTTTGGGAGATGTAAGAAATGTTCTTGCAGTTAAGGGTTAGTTTACAAT
```

Change region shown

Customize view

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Articles about the TP53 gene

[Association of polymorphic markers Arg72Pro of TP53 and T309g of MDM2 [Mol Biol (Mosk). 2014]

Ferroptosis as a p53-mediated activity during tumour suppression. [Nature. 2015]

Clinicopathological importance of Ki-67, p27, and p53 expression in gastric. [Turk J Med Sci. 2015]

See all...

Pathways for the TP53 gene

图形化界面显示

Graphics ▾

Send: ▾

Homo sapiens mRNA for P53, complete cds

GenBank: AB082923.1

[GenBank](#) [FASTA](#)

[Link To This Page](#) | [Feedback](#)



第一堂课内容提要

- 1. NCBI及GenBank的介绍
- 2. 如何检索
- 3. 解释检索结果
- 4. 下载选定结果
- 4. NCBI其他资源介绍

单个查询结果的序列下载

Display Settings: ☒ GenBank

Homo sapiens mRNA for P53, complete cds

GenBank: AB082923.1

[FASTA](#) [Graphics](#)

Go to: ☒

LOCUS AB082923 2451 bp mRNA line:
 DEFINITION Homo sapiens mRNA for P53, complete cds.
 ACCESSION AB082923
 VERSION AB082923.1 GI:23491728
 KEYWORDS .
 SOURCE Homo sapiens (human)
 ORGANISM [Homo sapiens](#)
 Eukaryota; Metazoa; Chordata; Craniata; Vertebrata;
 Mammalia; Eutheria; Euarchontoglires; Primates;
 Catarrhini; Hominidae; Homo.
 REFERENCE 1
 AUTHORS Azuma, K., Shichijo, S., Maeda, Y., Nakatsura, T., Ito, K., Koike, K. and Itoh, K.
 TITLE Mutated p53 gene encodes a nonmutated epitope recognized by HLA-B*4601-restricted and tumor cell-reactive CD8+ T lymphocytes
 JOURNAL Cancer Res. 63 (4), 854-858 (2003)
 PUBMED [12591737](#)
 REFERENCE 2 (bases 1 to 2451)
 AUTHORS Shichijo, S. and Itoh, K.
 TITLE Direct Submission
 JOURNAL Submitted (26-MAR-2002) Shigeaki Shichijo, Kurume Med. Sch., Dep. Immunol., 67-Asahi-machi, Kurume, Fukuoka, Japan (E-mail: shichijo@med.kurume-u.ac.jp, Tel: +81-942-31-7699, Fax: +81-942-31-7699)
 FEATURES
 source Location/Qualifiers
 ..2451

Send: ☒

Send: ☒

- ☒ Complete Record
- ☐ Coding Sequences
- ☐ Gene Features

Choose Destination

- ☒ File
- ☐ Clipboard
- ☐ Collections
- ☐ Analysis Tool

Download 1 items.

Format

- GenBank
- Summary
- GenBank
- GenBank (full)
- FASTA
- ASN.1
- XML
- INSDSeq XML
- TinySeq XML
- Feature Table
- Accession List
- GI List

Change region shown

Customize view

Analyze this sequence

Run BLAST

Pick Primers

Highlight Sequence Features

Find in this Sequence

Articles about the TP53 gene

[Association of polymorphic markers Arg72Pro of TP53 and T309g of MDM2 [Mol Biol (Mosk). 2014]

Ferroptosis as a p53-mediated activity during tumour suppression. [Nature. 2015]

Clinicopathological importance of Ki-67, p27, and p53 expression in gastric cancer. [Turk J Med Sci. 2015]

See all...

Pathways for the TP53 gene

TP53 Regulates Metabolic Genes

Transcriptional Regulation by TP53

Sphingolipid signaling pathway

[Association of polymorphic markers Arg72Pro of TP53 and T309g of MDM2 [Mol Biol (Mosk). 2014]

Ferroptosis as a p53-mediated activity during tumour suppression. [Nature. 2015]

Clinicopathological importance of Ki-67, p27, and p53 expression in gastric cancer. [Turk J Med Sci. 2015]

Batch Entrez



Batch Entrez

1

2

3

Use Batch Entrez to upload a file of GIs or accession numbers from the Nucleotide or Protein databases, or upload a list of record identifiers from other Entrez databases.

Tips : Some lists of record identifiers can be tens of thousands of lines long and Batch Entrez may not retrieve all records from one list. Split the list of identifiers into smaller files using a file splitting software or a file split command at the command prompt in UNIX or LINUX systems. Put several thousand record identifiers per file, left-formatted, and one per line. This may be necessary when you are loading large numbers of genome records. You can check the NCBI FTP site locations to download entire genome records. Also, use GIs rather than 'accession.version' numbers when making lists for batch entrez to fetch.

Please note that Batch Entrez will check for duplicate identifiers when reporting results from a list that you have imported.

When retrieving a list of Nucleotide accessions, you must select the specific component database from which the accessions or GIs were saved. For Nucleotide, choose either the CoreNucleotide, the EST or the GSS selection from the database menu. If you have a mixed list of nucleotide accessions or UIDs, you will need to run the Batch Entrez search three times. Select the database from the pull-down menu, CoreNucleotide, EST, and GSS separately.

In all cases, be certain to select the correct database for uploaded identifiers when using Batch Entrez, to ensure the expected records. For example, if you have saved a list of protein GIs, be sure to select the Protein database.

- Create a file with a list of GI or accession numbers and save it locally
- Select the database from which the list of accessions or UIDs originated
- Use the 'Browse' button to select the filename containing the list of UIDs from your system directory

<http://www.ncbi.nlm.nih.gov/sites/batchentrez>

```
Received lines: 50
Rejected lines: 0
Removed duplicates: 0
Passed to Entrez: 50
```

Retrieve records for 50 UID(s)

进入序列页面



NCBI Resources How To wang_ming_jie My NCBI Sign Out

Nucleotide Nucleotide Search Advanced Help

Species
Animals (50)
Customize ...

Molecule types
genomic DNA/RNA (36)
mRNA (14)
Customize ...

Source databases
GenBank (31)
RefSeq (6)
Customize ...

Sequence length
Custom range...

Release date
Custom range...

Revision date
Custom range...

[Clear all](#)

[Show additional filters](#)

Display Settings: ☒ Summary, 20 per page, Sorted by Default order [Send to:](#) ☒ [Filters:](#) [Manage Filters](#)

Results: 1 to 20 of 50 << First < Prev Page 1 of 3 Next > Last >>

- ☐ [Human mRNA fragment for phosphoprotein p53](#)
1. 2,066 bp linear mRNA
Accession: X01405.1 GI: 35215
[GenBank](#) [FASTA](#) [Graphics](#)
- ☐ [Human phosphoprotein p53 gene, exon 2](#)
2. 133 bp linear DNA
Accession: M22882.1 GI: 189465
[GenBank](#) [FASTA](#) [Graphics](#)
- ☐ [Human phosphoprotein p53 gene, exon 3](#)
3. 52 bp linear DNA
Accession: M22883.1 GI: 189466
[GenBank](#) [FASTA](#) [Graphics](#)
- ☐ [Human phosphoprotein p53 gene, exon 4](#)
4. 311 bp linear DNA
Accession: M22884.1 GI: 189467
[GenBank](#) [FASTA](#) [Graphics](#)
- ☐ [Human phosphoprotein p53 gene, exon 5](#)
5. 214 bp linear DNA

Results by taxon [Top Organisms \[Tree\]](#)
Homo sapiens (48)
Bubalus bubalis (1)
Canis lupus familiaris (1)

Find related data
Database:
[Find items](#)

Recent activity
[Turn Off](#) [Clear](#)

- Homo sapiens mRNA for P53, complete cds
Nucleotide
- Homo sapiens p53 (p53) gene, exon 7 and partial cds
Nucleotide
- human p53 (26102)
Nucleotide
- PubMed Help - PubMed Help

设置display setting



NCBI Resources How To wang_ming_jie My NCBI Sign Out

Nucleotide Nucleotide Search Help

Advanced

Species
Animals (50)
Customize ...

Molecule types
genomic DNA/RNA (36)
mRNA (14)
Customize ...

Source databases
GenBank (31)
RefSeq (6)
Customize ...

Sequence length
Custom range...

Release date
Custom range...

Revision date
Custom range...

Clear all
Show additional filters

Display Settings: Summary, 20 per page, Sorted by Default order

Format

- ☐ Summary
- ☐ GenBank
- ☐ GenBank (full)
- ☐ FASTA
- ☒ FASTA (text)
- ☐ ASN.1
- ☐ Revision History
- ☐ Accession List
- ☐ GI List

Items per page

- ☐ 5
- ☐ 10
- ☐ 20
- ☐ 50
- ☐ 100
- ☒ 200

Sort by

- ☒ Default order
- ☐ Accession
- ☐ Date Modified
- ☐ Date Released
- ☐ Organism Name
- ☐ Taxonomy ID

Send to: Last >>

Filters: Manage Filters

Results by taxon

Top Organisms [Tree]

- Homo sapiens (48)
- Bubalus bubalis (1)
- Canis lupus familiaris (1)

Find related data

Database: Select

Find items

Recent activity

Turn Off Clear

- Homo sapiens mRNA for P53, complete cds
Nucleotide
- Homo sapiens p53 (p53) gene, exon 7 and partial cds
Nucleotide
- human p53 (26102)
Nucleotide
- PubMed Help - PubMed Help

3. [Human phosphoprotein p53 gene, exon 3](#)
52 bp linear DNA
Accession: M22883.1 GI: 189466
[GenBank](#) [FASTA](#) [Graphics](#)

4. [Human phosphoprotein p53 gene, exon 4](#)
311 bp linear DNA
Accession: M22884.1 GI: 189467
[GenBank](#) [FASTA](#) [Graphics](#)

5. [Human phosphoprotein p53 gene, exon 5](#)
214 bp linear DNA

出现序列页面

```
>gi|35215|emb|X01405.1| Human mRNA fragment for phosphoprotein p53
CAAAACCTACCAGGGCAGCTACGGTTTCCGTCTGGGCTTCTTGCACTTCTGGGACAGCCAAGTCTGTGACT
TGCACGTACTCCCCTGCCCTCAACAAGATGTTTGGCAACTGGCCAAGACCTGCCCTGTGCAGCTGTGGG
TTGATTCCACACCCCGCCCGGCACCCGCGTCCGCGCCATGGCCATCTACAAGCAGTCACAGCACATGAC
GGAGGTTGTGAGGCGCTGCCCCACCATGAGCGCTGCTCAGATAGCGATGGTCTGGCCCCCTCTCAGCAT
CTTATCCGAGTGGAAAGGAAATTTGCGTGTGGAGTATTTGGATGACAGAAACACTTTTCGACATAGTGTGG
TGGTGCCTATGAGCCGCTGAGGTTGGCTCTGACTGTACCCACCATCCACTACAACATACATGTGTAACAG
TTCTGTCATGGGCGGCATGAACCGGAGGCCCATCTCACCATCATCACACTGGAAGACTCCAGTGGTAAT
CTACTGGGACGGAACAGCTTTGAGGTGCGTGTGTTGTCCTGTCTGGGAGAGACCGCGCACAGAGGAAG
AGAATCTCCGCAAGAAAGGGGAGCCTCACCAGAGCTGCCCCAGGGAGCACTAAGCGAGCACTGCCCAA
CAACACCAGCTCCTCTCCCCAGCCAAGAAAGAAACCACTGGATGGAGAATATTTACCCCTTCAGATCCGT
GGCGGTGAGCGCTTCGAGATGTTCCGAGAGCTGAATGAGGCCTTGGAACTCAAGGATGCCAGGCTGGGA
AGGAGCCAGGGGGGAGCAGGGCTCACTCCAGCCACCTGAAGTCCAAAAGGGTCAGTCTACCTCCCGCCA
TAAAAAATCATGTTCAAGACAGAAGGGCTGACTCAGACTGACATTCTCCACTTCTTGTTCCCACTGA
CAGCCTCCCCACCCCATCTCTCCCTCCCTGCGATTTTGGGTTTGGGTCTTTGAACCCCTGCTTGCAAT
AGGTGTGCGTCAGAAGCACCCAGGACTTCCATTTGCTTTGTCCCGGGCTCCACTGAACAAGTTGGCCTG
CACTGGTGTGTTGTTGTGGGAGGAGGATGGGAGTAGGACATACCAGCTTAGATTTAAGGTTTTTACT
GTGAGGGATGTTGGGAGATGTAAGAAATGTTCTTGCAATTAAGGGTTAGTTACAAATCAGCCACATTCT
AGGTAGGGGCCACTTCACCGTACTAACCAGGGAAGCTGTCCCTCACTGTTGAATTTTCTTAACCTCAA
GGCCCATATCTGTGAAATGCTGGCATTTGCACCTACCTCACAGAGTGCAATTTAGGGGTTAATGAAATAA
TGTACATCTGGCCTTGAACCACCTTTTATTACATGGGGTCTAGAACCTTGACCCCTTGAGGGTGCTTGT
TCCCTCTCCCTGTTGGTTCGGTGGGTTGGTAGTTTCTACAGTTGGGCAGCTGGTTAGGTAGAGGGAGTTGT
CAAGTCTCTGCTGGCCAGCCAAACCCGTCTGTGACAACCTCTTGGTGAACTTAGATCCTAAAAGGAAAT
GTACCCCATCCACACCCCTGGAGGATTTATCTCTTGATAGATGATCTGGATCCACCAAGACTTGTGTT
TAGCTCAGGGTCCAAATTTCTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTCTTTGAGACTGGGCTCTTTG
TTGCCCCAGGCTGGAGTGGAGTGGCGTATCTGGCTTACTGCAGCCTTTGCTCCCGGCTCGAGCAGTC
CTGCTCAGCCTCCGGAGTAGCTGGGACCACAGGTTCAATGCCACCATGGCCAGCCAACTTTTGCATGTTT
GTAGAGATGGGGTCTCAGAGTGTGGCCAGGCTGGTCTCAAACTCCTGGGCTCAGGCGATCCACCTGTC
TCAGCCTCCAGAGTGCTGGGATTACAATTTGTGAGCCACACGTCAGCTGGAAGGGTCAACATCTTTTA
CATCTGCAAGCACATCTGCATTTTACCCACCCCTTCCCTCTTCTCCCTTTTTATATCCCATTTTTAT
ATCGATCTCTTATTTTACAATAAAACCTTTGCTGCCA
```

```
>gi|189465|gb|M22882.1|HUMP53A02 Human phosphoprotein p53 gene, exon 2
TGGATCCTCTTGACGAGCCAGACTGCCTTCCGGGTCACTGCCATGGAGGAGCCGAGTCAGATCCTAGC
GTCGAGCCCCCTCTGAGTCAGGAAACATTTTCAGACCTATGGAACCTGTGAGTGGATCCATTG
```

```
>gi|189466|gb|M22883.1|HUMP53A03 Human phosphoprotein p53 gene, exon 3
GCTCTTGACTTTTCACTTCTGAAAAACAACGTTCTGGTAAGGACAAGGGTT
```

Ctrl-A 全选

Ctrl-C 复制

Ctrl-V 粘贴

任务完成！

或者采用“Send to”

NCBI Resources How To Sign in to NCBI

Nucleotide Nucleotide Search Advanced Help

Species Animals (67) Customize ...

Molecule types genomic DNA/RNA (67) Customize ...

Source databases RefSeq (67) Customize ...

Sequence length Custom range...

Release date Custom range...

Revision date Custom range...

Clear all Show additional filters

Summary 20 per page Sort by Default order

Items: 1 to 20 of 67

1. [Homo sapiens chromosome 15 sequence, ENCODE region ENr233](#)
500,000 bp linear DNA
Accession: NT_086346.3 GI: 42655550
[GenBank](#) [FASTA](#) [Graphics](#)

2. [Homo sapiens chromosome X sequence, ENCODE region ENm006](#)
1,338,447 bp linear DNA
Accession: NT_086362.5 GI: 84798482
[GenBank](#) [FASTA](#) [Graphics](#)

3. [Homo sapiens peptidylprolyl isomerase A pseudogene 29 \(PPIAP29\) on chromosome 6](#)
1,011 bp linear DNA
Accession: NG_007346.1 GI: 166091448
[GenBank](#) [FASTA](#) [Graphics](#)

Send to: Filters: [Manage Filters](#)

Choose Destination

☒ File ☐ Clipboard
☐ Collections

Download 67 items.

Format FASTA

Sort by Default order

Create File

Turn Off Clear

BRCA1[All Fields] AND biomol_genomic[PROP] AND "Hom Nucleotide

BRCA1[All Fields] AND biomol_genomic[PROP] AND "Hom Nucleotide

BRCA1[All Fields] AND

检索方式二： Blast



BLAST®

[Home](#)

BLAST finds regions of similarity between biological sequences. [more...](#)

New Try the new BLAST [homepage](#)

BLAST Assembled Genomes

Find Genomic BLAST pages:

GO

- ▣ [Human](#)
- ▣ [Mouse](#)
- ▣ [Rat](#)
- ▣ [Cow](#)
- ▣ [Pig](#)
- ▣ [Dog](#)

- ▣ [Rabbit](#)
- ▣ [Chimp](#)
- ▣ [Guinea pig](#)
- ▣ [Fruit fly](#)
- ▣ [Honey bee](#)
- ▣ [Chicken](#)

- ▣ [Zebrafish](#)
- ▣ [Clawed frog](#)
- ▣ [Arabidopsis](#)
- ▣ [Rice](#)
- ▣ [Yeast](#)
- ▣ [Microbes](#)

Basic BLAST

Choose a BLAST program to run.

[nucleotide blast](#)

Search a **nucleotide** database using a **nucleotide** query
Algorithms: blastn, megablast, discontinuous megablast

[protein blast](#)

Search **protein** database using a **protein** query
Algorithms: blastp, psi-blast, phi-blast, delta-blast

[blastx](#)

Search **protein** database using a **translated nucleotide** query

[tblastn](#)

Search **translated nucleotide** database using a **protein** query

[tblastx](#)

Search **translated nucleotide** database using a **translated nucleotide** query

- BLAST是一个NCBI开发的序列相似搜索程序，
- 还可作为鉴别基因和其遗传特点。
- BLAST能够在小于15秒的时间内对整个DNA数据库执行序列搜索。

<https://blast.ncbi.nlm.nih.gov/Blast.cgi>

BLAST® >> blastn suite

Standard Nucleotide BLAST

blastn blastp blastx tblastn tblastx

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ?

BLASTN programs search nucleotide databases using a nucleotide query. [more...](#)[Clear](#)

Query subrange ?

From

To

Or, upload file

Choose File

No file chosen ?

Job Title

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

☐ Human genomic + transcript ☐ Mouse genomic + transcript ☒ Others (nr etc.):

Nucleotide collection (nr/nt) ?

Organism

Optional

Enter organism name or id—completions will be suggested

☐ Exclude +

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown ?

Exclude

Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to

Optional

Entrez Query

Optional

☐ Sequences from type material

YouTube Create custom database

Enter an Entrez query to limit search ?

Program Selection

Optimize for

- ☒ Highly similar sequences (megablast)
☐ More dissimilar sequences (discontiguous megablast)
☐ Somewhat similar sequences (blastn)

BLAST

Search database Nucleotide collection (nr/nt) using Megablast (Optimize for highly similar sequences)

☐ Show results in a new window

解螺旋

整节课的问题

- 例如我对一个基因 **BRCA1**感兴趣，想知道：
 - 1) 基因在哪条染色体上？
 - 2) 该基因的功能是什么？
 - 3) 该基因的DNA序列是什么？ Protein序列是什么？

Summary

- NCBI 包含了许多的核酸序列信息
- 搜索引擎: Entrez
- 可通过关键词或者序列相似形来进行检索
- 重要的是对序列检索信息进行解读



thanks